- Interact with an agent using each type of memory store, and observe the differences.
- Compress the memory store, and analyze the effect on memory retrieval.

Summary

- Memory in AI applications differentiates between unstructured and structured memory, highlighting their use in contextualizing prompts for more relevant interactions.
- Retrieval augmented generation (RAG) is a mechanism for enhancing prompts with context from external documents, using vector embeddings and similarity search to retrieve relevant content.
- Semantic search with document indexing converts documents into semantic vectors using TF–IDF and cosine similarity, enhancing the capability to perform semantic searches across indexed documents.
- Vector databases and similarity search stores document vectors in a vector database, facilitating efficient similarity searches and improving retrieval accuracy.
- Document embeddings capture semantic meanings, using models such as OpenAI's models to generate embeddings that preserve a document's context and facilitate semantic similarity searches.
- LangChain provides several tools for performing RAG, and it abstracts the retrieval process, allowing for easy implementation of RAG and memory systems across various data sources and vector stores.
- Short-term and long-term memory in LangChain implements conversational memory within LangChain, distinguishing between short-term buffering patterns and long-term storage solutions.
- Storing document vectors in databases for efficient similarity searches is crucial for implementing scalable retrieval systems in AI applications.
- Agent knowledge directly relates to the general RAG pattern of performing question and answer on documents or other textual information.